

Lab 1: Use-Case Flow Specification

Problem Statement #23 — Hyperlocal Courier Dispatch & Tracking Engine

Use Case: UC-05 / UC-06 — Complete Delivery with OTP Verification

Primary Actor: Delivery Rider

Secondary Actor: Recipient

Preconditions

- The delivery request has been created, matched to a rider, and its status is "Out for Delivery".
- The rider has physically arrived at the drop-off location.
- The recipient has received a one-time password (OTP) via SMS/app notification, generated when the parcel was dispatched.

Postconditions

- The order status is updated to "Delivered", timestamped and geo-tagged with the rider's current GPS location.
- The sender client receives a real-time delivery confirmation notification.
- A proof-of-delivery record (OTP validation result, timestamp, location) is stored for the order.

Main Success Scenario

1. Rider arrives at the destination and taps "Complete Delivery" on the rider app.
2. System (UC-06, included) prompts the rider to enter the delivery OTP.
3. Recipient reads out the OTP received on their device and shares it with the rider.
4. Rider enters the OTP into the app.
5. System validates the entered OTP against the OTP generated for this order.
6. OTP matches; system accepts the entry.
7. System updates the order status to "Delivered" with a timestamp and the rider's GPS coordinates.
8. System sends a delivery-confirmation push notification to the sender client.
9. System stores the proof-of-delivery record.
10. Use case ends successfully.

Alternate Flow — 5a. OTP Mismatch

- 5a1. The entered OTP does not match the order's OTP.
- 5a2. System displays an "Invalid OTP, please try again" message and allows the rider to re-enter it.
- 5a3. If the OTP fails validation three times, the system locks OTP entry for 5 minutes, flags the order for support review, and displays a "Contact Support" option to the rider.
- 5a4. Rider either waits for the cooldown and retries from step 4, or contacts support to resolve the mismatch; the use case resumes at step 4 once a valid OTP is entered, or ends in failure if support cancels the order.